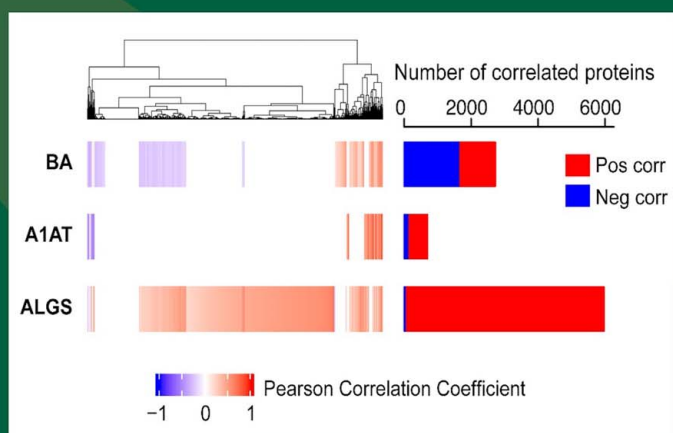


Plasma proteome correlations with liver stiffness in pediatric cholestasis implicate epithelial to mesenchymal transition

VISUAL ABSTRACT

Plasma proteome correlations with liver stiffness in pediatric cholestasis implicate epithelial to mesenchymal transition



Correlation between protein abundance and liver stiffness measurement in biliary atresia (BA), alpha-1 antitrypsin deficiency (A1AT) and Alagille syndrome (ALGS)

- Of > 7,000 proteins quantified by modified aptamer scanning, multiple correlations were observed with LSM in a disease-specific manner
- LSM and clinical outcome can be predicted by protein abundance
- Liver stiffness-linked proteins may originate from macrophage, mesothelial, mesenchymal and endothelial liver cells in BA and A1AT
- Pathway analyses suggest a critical role of epithelial to mesenchymal transition in hepatic fibrogenesis

ORIGINAL ARTICLE

OPEN

Plasma proteome correlations with liver stiffness in pediatric cholestasis implicate epithelial to mesenchymal transition

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Abbreviations: A1AT, alpha-1 antitrypsin deficiency; ALGS, Alagille syndrome; BA, biliary atresia; ChILDRn, Childhood Liver Disease Research Network; C-index, concordance index; dCEPH, definite clinically evident portal hypertension; EMT, epithelial-to-mesenchymal transition; FDR, false discovery rate; FORCE, Fibroscan in Pediatric Cholestatic Liver Disease; INR, international normalized ratio; IPA, ingenuity pathway analysis; IRB, Institutional Review Board; KNN, k-nearest neighbor; LSM, liver stiffness measurement; LT, liver transplantation; LTFS, liver transplant free survival; MCC, maximal clique centrality; ML, machine learning; ORA, over-representation analysis; pCEPH, possible clinically evident portal hypertension; RF, random forest; RMSE, root mean squared error; SVM, support vector machine; VCTE, vibration-controlled transient elastography; WGCNA, weighted gene correlation network analysis.

Benjamin L. Shneider and Rupa S. Kanchi are co-first authors.

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Abstract

Background: Pediatric cholestatic liver diseases can be characterized by rapidly progressive fibrosis. A multicenter cross-sectional analysis of vibration-controlled elastography in biliary atresia (BA), alpha-1 antitrypsin deficiency (A1AT), and Alagille syndrome (ALGS) was leveraged to interrogate the plasma proteome relative to liver stiffness measurements (LSM).

Methods: Slow off-rate modified aptamer scanning profiling of >7000 proteins in plasma from 187 children with BA (n=93), A1AT (n=31), ALGS (n=46), and healthy pediatric controls (n=17) was performed, and correlations with LSM were undertaken.

Results: There was an abundance of LSM correlated proteins (BA n=2720, A1AT n=694, ALGS n=5968). Interestingly, a distinct plasma proteome was found in ALGS relative to BA and A1AT. Weighted Correlation Network Analysis identified groups of proteins with strong LSM correlation (eg, in a BA module of interest, Pearson correlation coefficient 0.79, p=5.0-21). Machine learning developed models predicting LSM as a continuous variable (median R²=0.62 for BA). For BA, time to transplant could be predicted equally well by the proteome or clinical parameters (elastic net models achieved a C-index using proteome 0.91, clinical parameters 0.91, proteome and clinical parameters 0.90). Single-cell transcriptomics predicted the potential hepatic cell of origin for the most informative proteins, which included macrophage, mesenchymal, mesothelial, and endothelial cells. The epithelial-to-mesenchymal transition pathway was enriched in LSM correlated proteins in all 3 diseases.

Conclusions: The plasma proteome is highly correlated in a disease-specific fashion with LSM in BA, A1AT, and ALGS. These correlations provide unique opportunities to identify biomarkers and focus attention on epithelial-to-mesenchymal transition in pediatric cholestasis.

Keywords: Alagille syndrome, alpha-1 antitrypsin, biliary atresia, biomarker, cirrhosis, elastography

INTRODUCTION

Cholestatic diseases constitute the majority of the diagnoses leading to pediatric liver transplantation. Advancing fibrosis and complications of portal hypertension underlie most of the indications for transplant in this population. Often these complications are in the context of relatively intact hepatic function. As such, an accurate assessment of the severity and pathophysiologic underpinnings of hepatic fibrosis in pediatric cholestasis is paramount. The etiopathogenesis of pediatric portal hypertension and fibrosis is distinct from that seen in adults and necessitates pediatric-specific investigations.^[1] Many consider histopathologic assessment of fibrosis to

be the gold standard, although the patchy nature of fibrosis and the invasiveness of liver biopsy in children challenge its utility in pediatric liver diseases.^[2]

There is a critical and unmet need to develop noninvasive markers of both hepatic fibrosis and liver disease severity in pediatrics. A fundamental problem in addressing this issue is the relative unavailability of a valid reference standard to use in the development of noninvasive markers. Liver biopsy data in pediatrics are relatively scant, typically available at diagnosis and transplant. Surveillance biopsies are generally not performed in pediatrics, as there is uncertainty about the risk-benefit of the procedure in routine clinical care. There is also a lack of consensus on a global marker of

disease severity. Disease-specific differences in natural history and relevant clinical metadata complicate the goal. Noninvasive measurement of liver stiffness holds promise as a reference standard for the development of biomarkers. Liver stiffness measurement (LSM), as assessed by vibration-controlled transient elastography (VCTE), correlates with features of portal hypertension and measures of liver disease severity in children with biliary atresia (BA), alpha-1 antitrypsin deficiency (A1AT), Alagille syndrome (ALGS), and predicts outcome in BA.^[3,4]

Noninvasive biomarkers of liver disease severity may be discovered using elastographic assessment of liver stiffness as the reference standard. A targeted proteomic approach using liver stiffness as the reference standard has identified a number of potential biomarkers.^[5] Methodologies for analyzing the proteome have matured and provide tremendous opportunities for discovery.^[6] The slow off-rate modified aptamer scan (SomaScan, SomaLogic, Inc.) was used to identify several serum biomarkers that associated with clinical features of portal hypertension in children with BA.^[7] The power of this approach is illustrated by the ability of the serum proteome to correlate not only with multiple features of the liver biopsy in metabolic dysfunction associated steatotic liver disease, but to identify novel signals of response to drug interventions.^[8] The current study leverages VCTE LSM assessment of liver stiffness conducted by the NIDDK/NIH-supported Childhood Liver Disease Research Network (ChILDRen) to identify plasma proteins that correlate with liver disease severity in children with BA, A1AT, and ALGS.

METHODS

Sample selection

Analyses were performed on 170 plasma samples derived from participants enrolled in the Fibroscan in Pediatric Cholestatic Liver Disease (FORCE) study, a prospective analysis of VCTE.^[3] Written informed consent was obtained from caregivers or the participant, and assent was obtained from the child when appropriate according to local Institutional Review Board (IRB) rules. This study was approved by local and central IRBs and complied with the Declarations of Helsinki and Istanbul and Good Clinical Practice Guidelines. Salus IRB served as the central IRB (#16096) and the analysis of the proteome was approved by the Baylor College of Medicine IRB (#H-48308). The study was registered at ClinicalTrials.gov (NCT 02922751). Plasma samples were used from participants with a valid baseline LSM and who had relevant clinical parameters [eg, laboratory testing including total bilirubin, ALT, AST, albumin, international normalized ratio (INR), and platelet count] associated with their LSM.

Samples were selected to represent the enrollment in FORCE (50% BA, 25% ALGS, and 25% A1AT) and to provide a spectrum of liver disease severity as determined by clinically evident portal hypertension (CEPH) status.^[9] Given the limited prevalence, all participants with possible or definite CEPH (pCEPH or dCEPH) in the ALGS and A1AT cohorts were selected for this analysis. For BA, equal numbers of absent, possible, and definite CEPH were utilized. Resource limitations limited the analysis to plasma from 170 of the 330 participants in FORCE who had valid baseline LSM and available plasma samples and precluded the inclusion of normal controls.^[5] Liver histology was not available at the baseline visit in FORCE. An overview of the analytical approaches described below is presented in Figure 1. Time to transplant (n = 15) or death (n = 0) in participants with BA was derived from previously published analyses.^[4]

Healthy control participants (n = 17) were recruited from a general pediatric gastroenterology clinic, in patients without any known liver diagnoses who were being evaluated for constipation. These participants did not undergo LSM or the routine laboratory testing performed in the FORCE participants. This cohort was recruited to provide control values for the proteomic analyses. This study was conducted in accordance with the Declarations of Helsinki and Istanbul and was approved by the Institutional Review Board at the University of Southern California (CHLA-19-00177). Normalization of the SomaScan proteomic data was performed as previously reported.^[10]

SomaScan proteomics—differential expression

Proteomics profiling for over 7000 proteins was conducted on plasma samples using the SomaScan platform.^[11] SomaScan probes were annotated with the R package SomaDataIO. Differential protein expression across the 3 diseases was evaluated using an ANOVA-based *F*-statistic. Pairwise differential protein expression between diseases was evaluated using moderated *t*-statistics implemented using limma. Multiple hypothesis testing correction was performed using the Benjamini–Hochberg method (false discovery rate —FDR), and proteins were considered differentially expressed when FDR < 0.05.

Over-representation analysis

The Hallmark gene set collection was utilized to perform over-representation analysis (ORA) to investigate pathway enrichment in protein. ORA was conducted using a hypergeometric test, with significance achieved at FDR < 0.05.

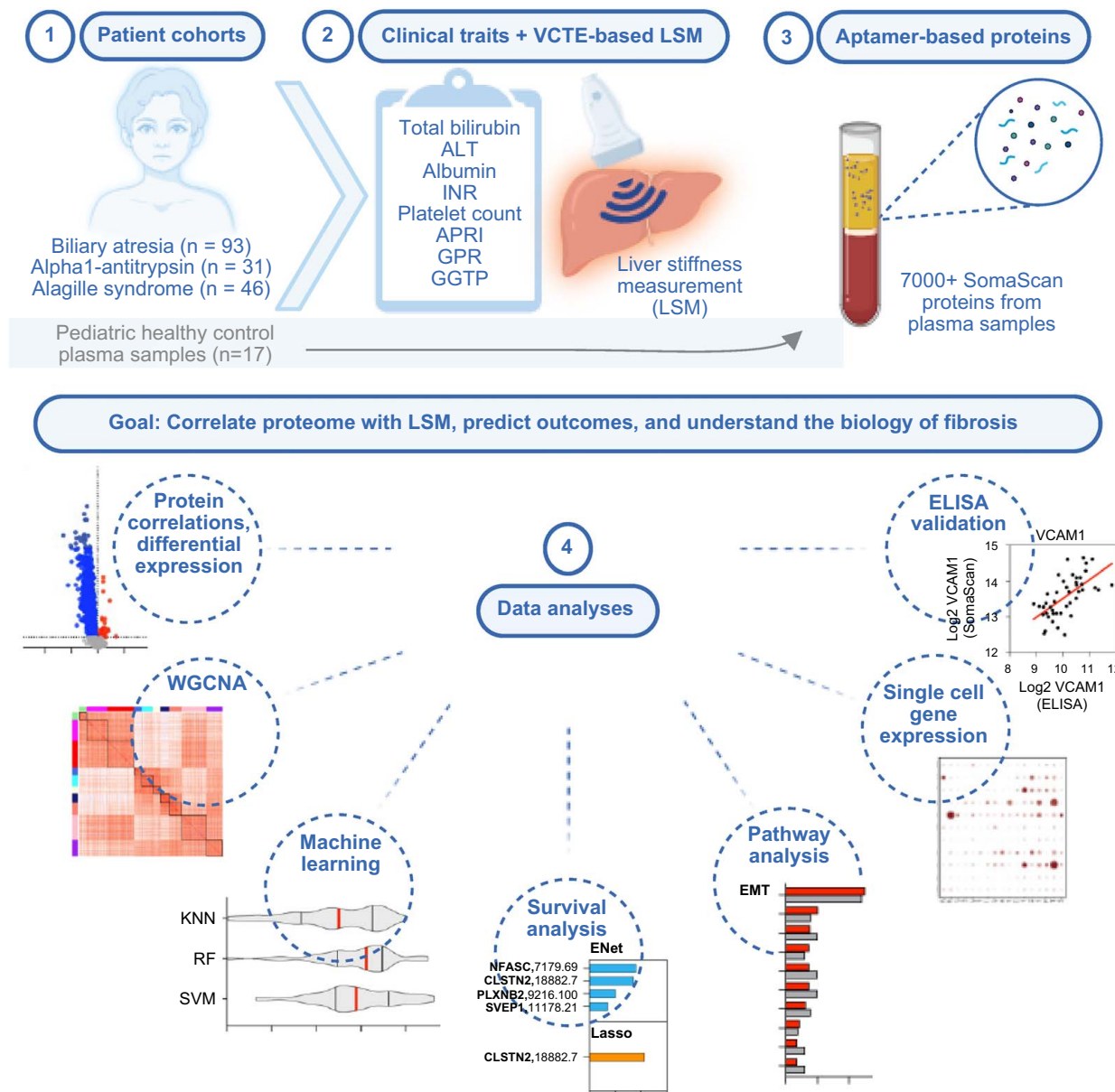


FIGURE 1 Analytical approach overview. Blood samples were collected from participants with biliary atresia (BA), alpha-1 antitrypsin deficiency (A1AT), and Alagille syndrome (ALGS); liver stiffness measurement (LSM) was assessed using VCTE; protein expression was measured using the SomaScan aptamer-based platform. A cohort of healthy pediatric participants evaluated for constipation without evident liver disease was recruited as normal pediatric controls for the proteomic analyses. Protein expression was correlated with LSM. Network analysis was conducted for each disease using WGCNA. LSM was predicted using machine learning regression. Informative proteins were characterized by enriched pathways, protein–protein interaction networks, and cell-type annotation. The expression of selected proteins was confirmed by ELISA. Association of proteins and clinical features with liver transplant outcome was evaluated using survival analysis. Created in BioRender <https://BioRender.com/47unsns>. Abbreviations: APRI, aspartate aminotransferase to platelet ratio index; GGTP, gamma-glutamyl transpeptidase; GPR, gamma-glutamyl transpeptidase to platelet ratio; INR, international normalized ratio; KNN, k-nearest neighbor; RF, random forest; SVM, support vector machine; VCTE, vibration-controlled transient elastography; WGCNA, weighted gene correlation network analysis.

Correlation of individual proteins with liver stiffness measurement

For each disease cohort, Pearson correlation was computed between each protein and LSM. Significant

protein associations, at FDR < 0.25, and their overlaps were plotted using the ComplexHeatmap package. ORA was performed to identify pathways enriched in LSM-correlated proteins for each cohort, with significance achieved at FDR < 0.05.

ELISA validation of SomaScan quantification

Residual plasma from 75 participants was analyzed by ELISA for VCAM1 (Thermo Fisher Scientific [KHT0601]) and FRZB (Abcam [ab267653]) following manufacturer instructions.

Network analysis

Weighted correlation network analysis (WGCNA) was performed using WGCNA to identify co-expressed protein modules.^[12] For LSM-associated protein modules, identified using Pearson correlation at p -value < 0.05 , protein–protein interaction networks were constructed using the STRING database.^[13] Hub proteins were identified and ranked by maximal clique centrality (MCC) measure computed by the cytoHubba plugin.^[14] Protein interaction networks, with nodes annotated by their MCC ranks, were visualized using the Cytoscape platform.^[15] Enriched Hallmark pathways in each module were determined using ORA as described above.

Qiagen Ingenuity Pathway Analysis

Canonical pathway enrichment and network plots were generated using Qiagen Ingenuity Pathway Analysis (IPA),^[16] using proteins correlated with LSM. A right-tailed Fisher exact test was used to compute the enrichment significance. Network plots show molecular interactions, with molecules depicted as nodes and their biological relationships as edges. Each edge is supported by at least one reference from the Qiagen knowledge base.

Machine learning regression

Proteins predictive of LSM were identified for each disease cohort using machine learning. Data were partitioned 50 times into training and testing sets, with 80% of data used for model training and 20% for testing. Regression models were built using random forest (RF),^[17] support vector machines (SVMs), and k-nearest neighbor (KNN) methods implemented in the caret package. Model performance was assessed using root mean squared error (RMSE) and squared Pearson correlation coefficient (R^2). For each method, proteins informative in at least 50% of cross-validation iterations were selected, then ranked using their feature importance score. ggplot2 was used to plot the rank concordance of selected informative proteins across the 3 methods.

Liver transplant outcome analysis

We modeled liver transplant-free survival (LTFS) in the BA cohort ($n = 93$) using 4 sets of predictor variables: (1) proteins, (2) clinical variables without LSM: total bilirubin, ALT, albumin, INR, platelet count, APRI (aspartate aminotransferase to platelet ratio index), GPR (gamma-glutamyl transpeptidase to platelet ratio), and GGTP (gamma-glutamyl transpeptidase), (3) clinical variables including LSM, and (4) proteins and clinical variables. Data were partitioned 50 times into 70% training and 30% testing sets. For each iteration and predictor set, (1) regularized Cox models using lasso and elastic net were implemented; non-zero coefficients from these models, representing predictor impact on the HR, were used to rank the predictors, (2) random survival forests with 500 trees were grown. Log-rank statistic was used as a splitting rule, and predictor importance to the model was assessed by permutation testing. Random forest survival analysis and regularized Cox models were implemented using R packages ranger and glmnet.^[17,18] Model performance was evaluated using the test set concordance index (C-index).

Integration with single-cell transcriptomics from a liver model of cirrhosis

Non-parenchymal cirrhotic and healthy liver single-cell expression was downloaded from GEO GSE136103 with clinical parameters for adults with cirrhosis as described.^[18] Data were processed using Scanpy,^[19] and cell types annotated using known markers. Dotplots of selected genes across all cell types were generated using Seurat.^[20]

RESULTS

Clinical cohort description

Samples were analyzed from 93 FORCE participants (average age 9.9 y) with BA, 31 with A1AT, and 46 with ALGS, and an additional 17 healthy controls (average age 11.8 y). Clinical characteristics and demographics of the FORCE cohorts are presented in Figure 2 and Supplemental Table S1, <http://links.lww.com/HC9/C98>. Participants had preserved hepatic synthetic function characterized primarily by normal albumin (median 4.4, 4.3, and 4.3 A1AT, ALGS, and BA) and INR (median 1.10, 1.04, and 1.10 A1AT, ALGS, and BA). Thrombocytopenia as a manifestation of portal hypertension was most prevalent in BA, where 35.5% of participants had a platelet count less than 150,000 (22.6% A1AT and 19.6% ALGS). The median liver stiffness was 7, 9, and 10 kPa for A1AT, ALGS, and BA.

Distinct proteomics profiles for BA and A1AT compared with ALGS

A clustered heatmap based on the 1699 proteins differentially expressed ($FDR < 0.05$) across the 3 disease cohorts revealed a distinct pattern of protein abundance in ALGS relative to BA and A1AT (Figure 3A and Supplemental Table S2, <http://links.lww.com/HC9/C99>). The majority of differentially expressed proteins were found between ALGS and A1AT ($n = 4237$) or ALGS and BA ($n = 2246$), including 2087 common to BA and A1AT proteins that were differentially expressed compared with ALGS (Figures 3B, C). Twenty-five proteins were found to be different

between each pair of diseases, BA, ALGS, and A1AT (Figure 3C). As expected, plasma alpha-1 antitrypsin levels were markedly reduced in A1AT (SERPINEA1, Figure 3D).

Plasma protein levels correlate with LSM in a disease-specific manner

A large number of proteins correlated with LSM ($FDR < 0.25$); the correlations were disease-specific (Figure 4A). In ALGS, the correlations were primarily positive, while the patterns were mixed for BA and A1AT. A distinct set of 3578 proteins had a positive correlation

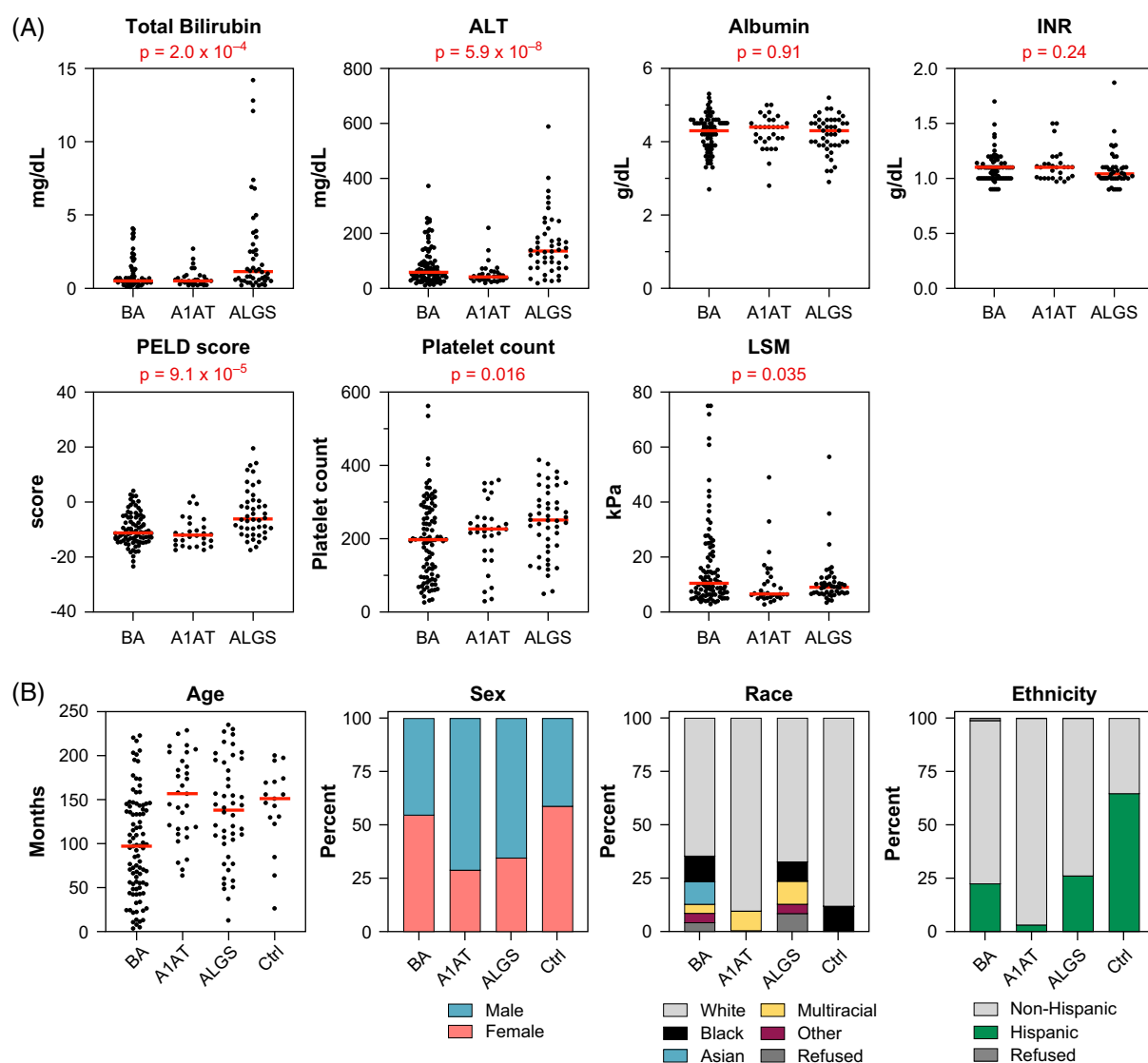


FIGURE 2 Clinical and demographic parameters. (A) Scatterplots for clinical parameters with median depicted by a horizontal red line and p -value for difference among the 3 diagnoses (ANOVA). (B) Demographic parameters are depicted for the 3 diagnoses and the healthy pediatric controls (Ctrl). Abbreviations: A1AT, alpha-1 antitrypsin deficiency; ALGS, Alagille syndrome; BA, biliary atresia; INR, international normalized ratio; LSM, liver stiffness measurement; Pediatric End-Stage Liver Disease.

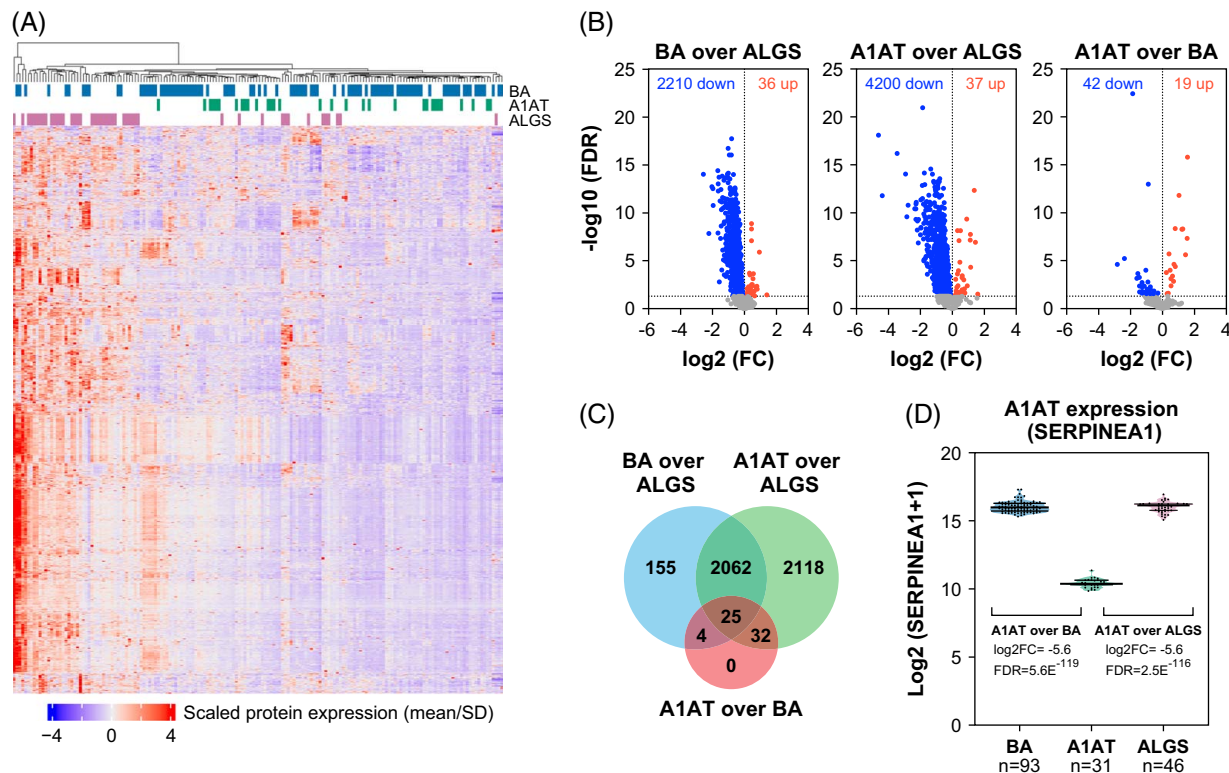


FIGURE 3 Global patterns of the plasma proteome. (A) Clustered heatmap of differentially expressed proteins (ANOVA) across the 3 diseases. (B) Differential protein expression analysis for pairwise comparison among the 3 diseases. (C) Overlap of differentially expressed proteins. (D) A1AT expression in the 3 diseases. Abbreviations: A1AT, alpha-1 antitrypsin deficiency; ALGS, Alagille syndrome; BA, biliary atresia; $\log_2 \text{FC}$, $\log_2$ fold change.

with LSM in ALGS (Figure 4B). Interestingly, a set of 1269 proteins was identified with opposite correlation directions in BA and ALGS (Figure 4B). Hallmark pathway analysis of these proteins revealed enrichment of PI3-AKT-mTOR signaling, MYC targets, mTORC1 signaling, and fatty acid metabolism. The upset plot further discovered 262 proteins positively correlated with LSM and 17 proteins negatively correlated with LSM common across the 3 diseases (Figure 4B); these proteins were enriched for epithelial-to-mesenchymal transition (EMT), apical junction, and coagulation pathways.

Hallmark pathway enrichment of proteins either positively or negatively correlated with LSM in each disease model revealed commonly enriched pathways (Figure 4C). In particular, the EMT pathway was the most enriched in proteins correlated with LSM in all 3 diseases. Pathway enrichment performed separately in positively and negatively associated proteins revealed that EMT was enriched in positively correlated proteins in all disease models; however, coagulation and complement pathways were enriched by proteins with a negative correlation with LSM in A1AT and ALGS and with a positive correlation in BA (Supplemental Figure S1, <http://links.lww.com/HC9/C100>).

Plasma proteomics relative to healthy children

A large number of proteins were differentially expressed relative to healthy children in BA (n=1040), A1AT (n=832), or ALGS (n=2391) (Supplemental Figure S2, <http://links.lww.com/HC9/C100>). Similarly, a large number of proteins were differentially expressed in participants with BA (n=1560), A1AT (n=691), or ALGS (1814) who had an LSM > 15, which has been associated with worse outcome (Supplemental Figure S2, <http://links.lww.com/HC9/C100>).^[4] There was a large overlap between proteins correlated with LSM and those that were differentially expressed relative to healthy children (Supplemental Figure S2, <http://links.lww.com/HC9/C100>). Proteins positively correlated with LSM overlapped with proteins with increased expression in children with diseases relative to healthy children; conversely, proteins negatively correlated with LSM overlapped with proteins with lower expression in disease relative to health. The number of proteins in overlap was higher in those with an LSM > 15 in BA and A1AT (BA—825 vs. 442, A1AT—301 vs. 140, ALGS—1436 vs. 2006). MMP-7, ENPP-7, and PIGR were elevated in BA with high LSM compared with healthy children, as has been previously described.^[21,22] Of the differentially

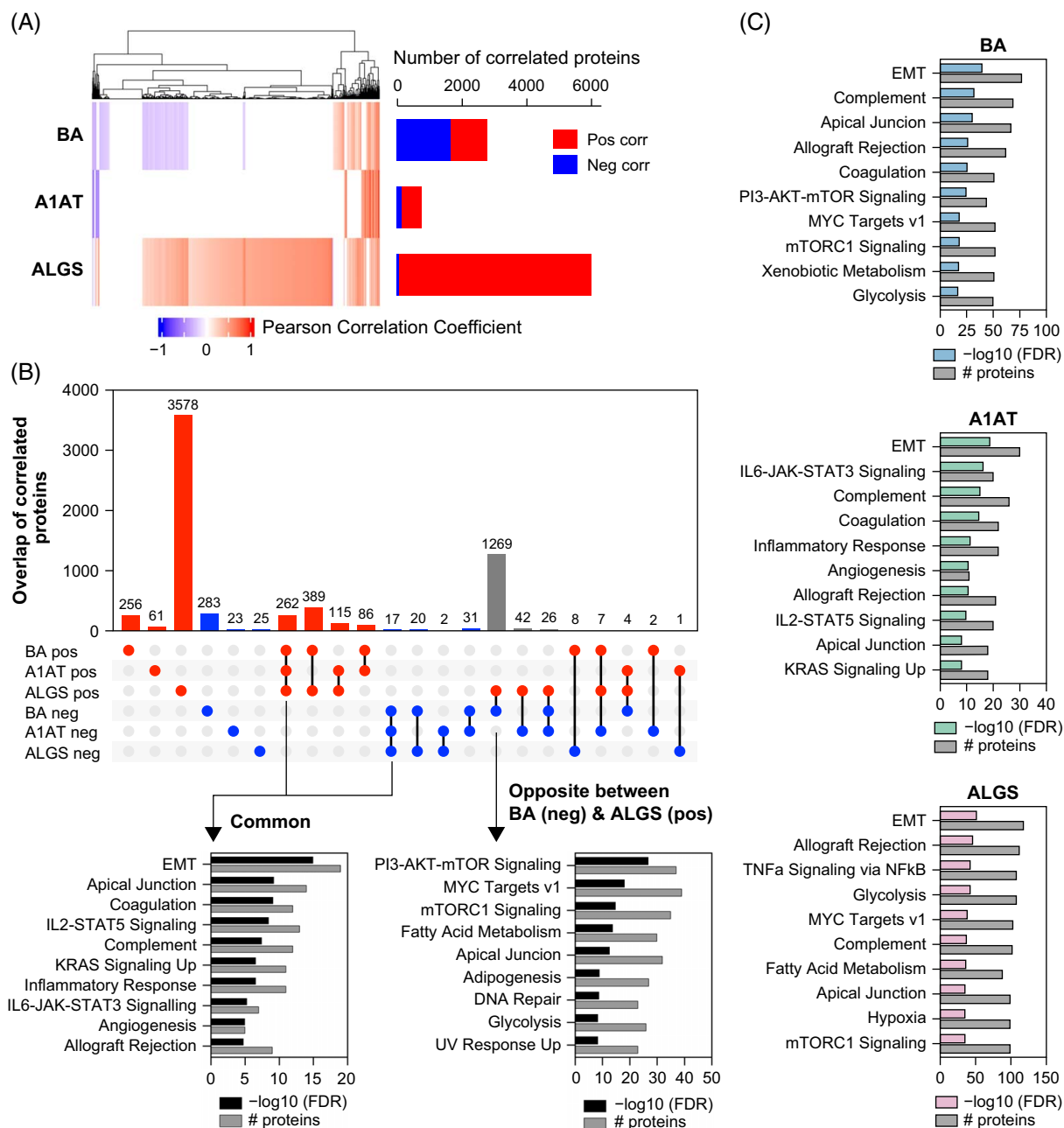


FIGURE 4 Correlation analysis between proteins and LSM. (A) Correlations between proteins and LSM in the 3 diseases. (B) Disease and direction-specific overlaps of proteins correlated with LSM. Top 10 Hallmark pathways for (1) disease-independent directionally uniform correlated proteins (n = 279) and (2) protein correlated in opposite directions between BA and ALGS (n = 1269). (C) Top 10 Hallmark pathways enriched in proteins correlated with LSM independent of direction of correlation. Abbreviations: A1AT, alpha-1 antitrypsin deficiency; ALGS, Alagille syndrome; BA, biliary atresia; EMT, epithelial-to-mesenchymal transition; FDR, false discovery rate; LSM, liver stiffness measurement.

expressed proteins, there were more that were unique to ALGS compared with BA or A1AT (Supplemental Figure S3, <http://links.lww.com/HC9/C100>).

Network analysis of plasma proteomics and liver stiffness

WGCNA identified 14 modules of co-expressed proteins in BA that had variable overall correlation (−0.26 to

0.79) with LSM and consisted of 28 to 3589 proteins per module (Figure 5A and Supplemental Table S3, <http://links.lww.com/HC9/C101>). The BA-pink module with 92 proteins is most significantly associated with LSM (ME_{pink} × LSM correlation coefficient 0.79, $p = 5 \times 10^{-21}$, Figure 5B). The top 20 pink module proteins based on LSM correlation are listed in Figure 5C. VWF*, MET*, VCAM1*, CSF1R*, and BMP4 were identified as prominent hub proteins, ranked by MCC measure, in the pink module protein–

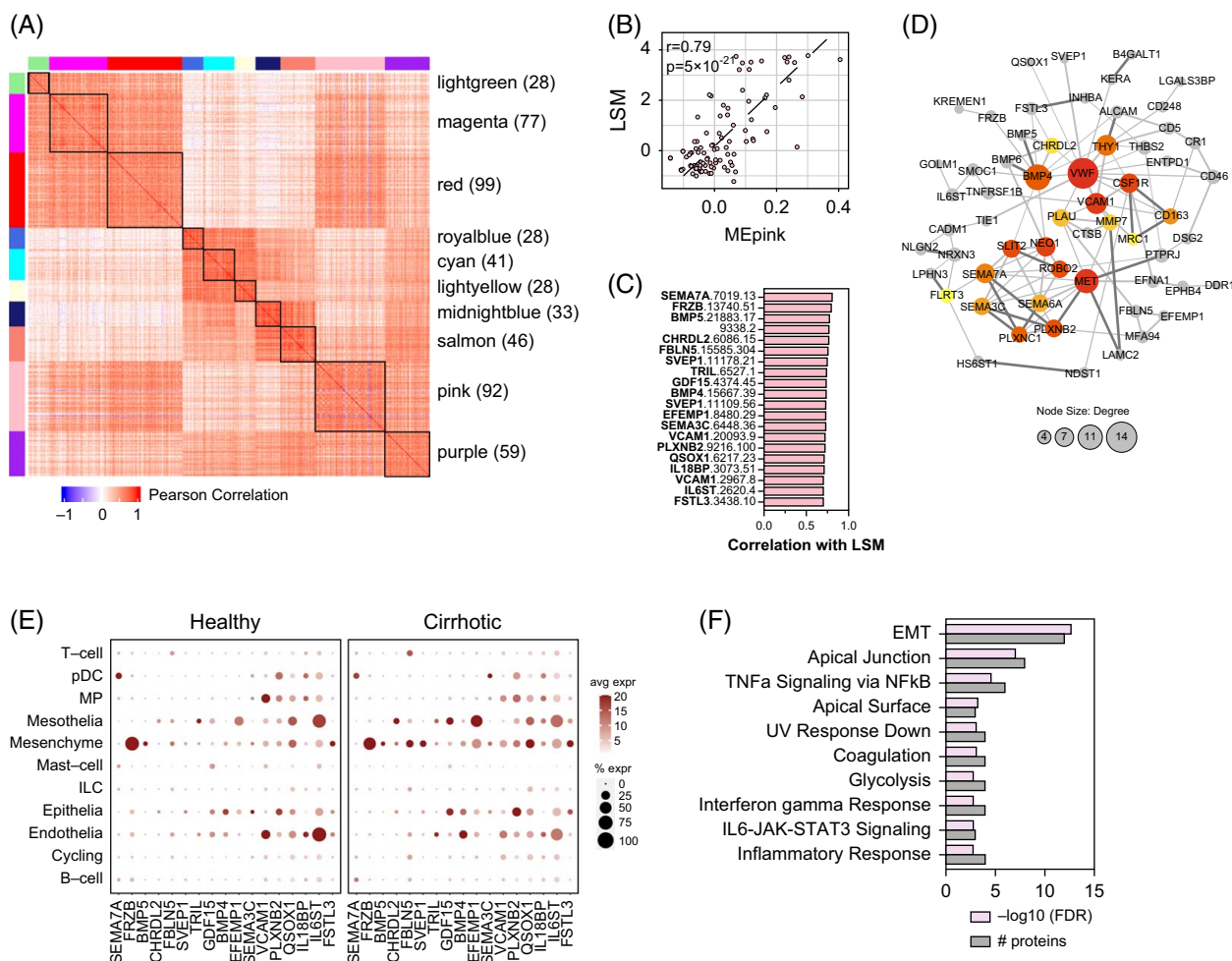


FIGURE 5 Gene correlation network analysis in BA. (A) Correlation heatmap of WGCNA modules of co-expressed proteins. (B) Scatterplot of LSM and the eigengene scores for the WGCNA pink module with 92 proteins. (C) Top 20 pink module protein LSM correlations. (D) Protein-protein interaction network of pink module proteins. (E) Gene expression in single-cell RNA-seq cell types for the top 20 pink module LSM-associated proteins. (F) Top 10 enriched Hallmark pathways in the pink module. Abbreviations: BA, biliary atresia; EMT, epithelial-to-mesenchymal transition; FDR, false discovery rate; LSM, liver stiffness measurement; WGCNA, weighted gene correlation network analysis.

protein interaction network (Figure 5D) (proteins noted by * were significantly different in participants with LSM > 15 compared with healthy children with representative examples shown in Supplemental Figure S4, <http://links.lww.com/HC9/C100>). Identification of the hepatic cell of origin for these plasma proteins is relevant to understanding the biology of liver stiffness and presumed underlying fibrosis. Specific cell types were derived from published studies of nonparenchymal cells in human liver, both normal and cirrhotic, using data from GEO GSE136103 and annotated using published cell-specific markers (Supplemental Figure S5, <http://links.lww.com/HC9/C100>).^[18] Genes encoding the top 20 LSM-associated proteins from the pink module are expressed preferentially in mesothelial (GDF15*, EFEMP1*, QSOX1*, IL6ST*), mesenchymal (FRZB*, FBLN5*, EFEMP1*, QSOX1*, FSTL3*), epithelial (GDF15*, BMP4, PLXNB2), and endothelial (BMP4, VCAM1*, IL6ST*) cells (Figure 5E). In cirrhotic versus normal liver, FBLN5*, SVEP1*, and EFEMP1* are

increased in mesenchymal cells, GDF15* is increased in epithelial cells, and BMP4 is increased in endothelial cells. EMT is the top significantly enriched Hallmark pathway with 12 genes overlapping the pink module (Figure 5F). IPA of the pink module proteins identified axonal guidance signaling as the top canonical pathway (Supplemental Table S4, <http://links.lww.com/HC9/C102>). Protein abundance of FRZB* and VCAM1*, assessed by ELISA, correlated with SomaScan values for VCAM1* and FRZB* (Supplemental Figure S6A, <http://links.lww.com/HC9/C100>) and was strongly associated with LSM (Supplemental Figure S6B, <http://links.lww.com/HC9/C100>).

WGCNA was conducted in A1AT and ALGS as well. Additional modules from BA, and modules from A1AT and ALGS that strongly correlate with LSM are shown in Supplemental Figures S7–S9, <http://links.lww.com/HC9/C100>. The proteins in the BA red module are derived predominantly from mesenchymal, mesothelial, and endothelial cells and are

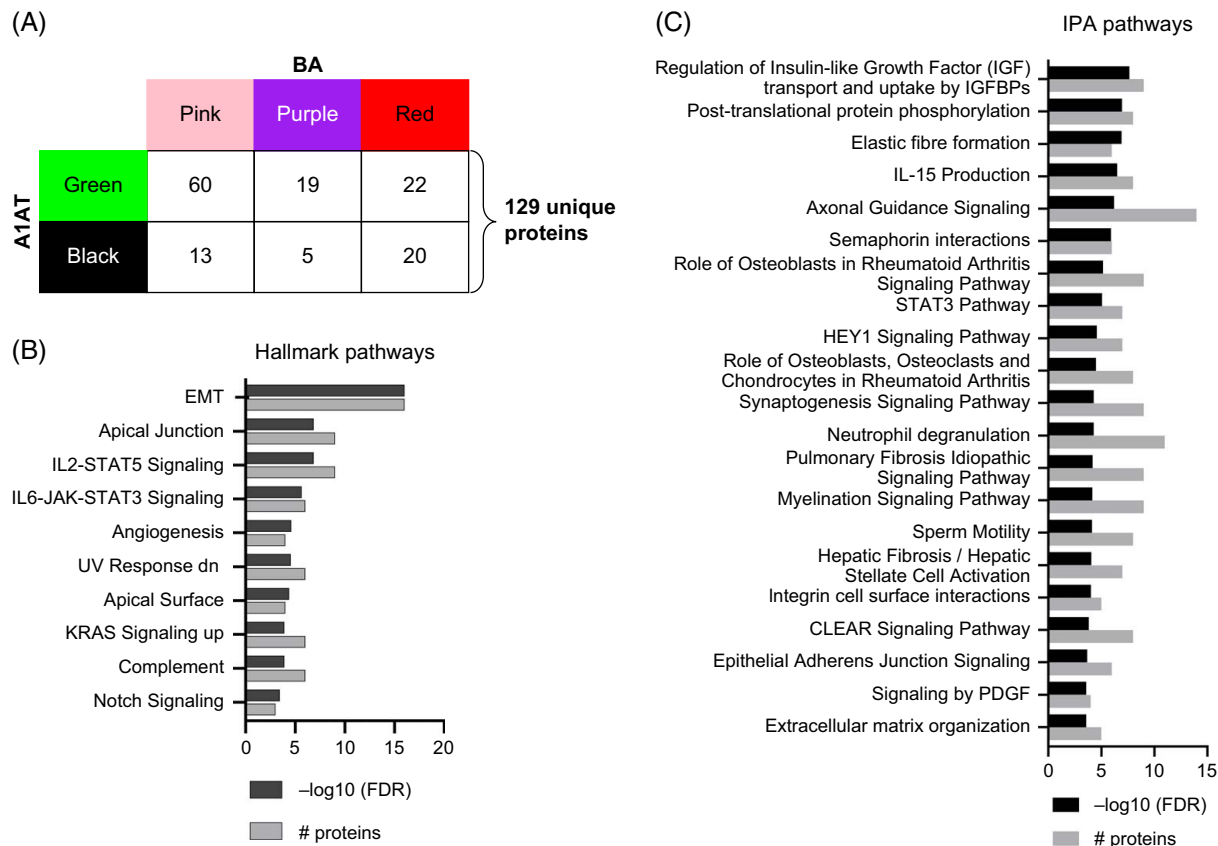


FIGURE 6 Protein overlap of WGCNA modules correlated with LSM in BA and A1AT. (A) Protein overlap among the LSM-correlated pink, purple, and red modules from BA and the green and black modules from A1AT. (B) The 129 unique overlapping proteins. (C) Top 10 enriched Hallmark pathways. (C) Top 20 enriched IPA pathways. Abbreviations: A1AT, alpha-1 antitrypsin deficiency; IPA, Ingenuity Pathway Analysis; EMT, epithelial-to-mesenchymal transition; FDR, false discovery rate; LSM, liver stiffness measurement; WGCNA, weighted gene correlation network analysis.

enriched in the EMT pathway (Supplemental Figure S7, <http://links.lww.com/HC9/C100>). The green and black modules for A1AT have similarities to the findings in BA (Supplemental Figure S8, <http://links.lww.com/HC9/C100>). The top proteins from the LSM-associated modules for ALGS were distinct from BA and A1AT and included a wide range of cell types enriched in different Hallmark pathways (Supplemental Figure S9, <http://links.lww.com/HC9/C100>). Given the similarities in patterns of correlation between LSM and the plasma proteome in BA and A1AT, analyses were performed on the overlapping proteins for these 2 diseases (BA—pink, purple, and red; A1AT—green and black; Figure 6). Overall, 129 proteins demonstrated a common positive correlation with LSM, and their top-enriched Hallmark pathway was EMT. IPA revealed enrichment of axonal guidance signaling along with hepatic fibrosis/hepatic stellate cell activation (Figure 6). Examination of the axonal guidance signaling pathway reveals proteins primarily associated with cellular membranes (Supplemental Figure S10, <http://links.lww.com/HC9/C100>).

Machine learning to predict LSM

Three machine learning approaches [random forest (RF), support vector machine (SVM), and k-nearest neighbor (KNN)] were used to determine proteins that could predict LSM as a continuous variable (BA—Figure 7, A1AT and ALGS—Supplemental Figures S11, S12, <http://links.lww.com/HC9/C100>). Predictive performance was higher for BA ($R^2=0.62$) and A1AT ($R^2=0.75$), compared with a lower performance in ALGS ($R^2=0.21$). The top informative proteins for BA from the RF model are shown along with the direction of correlation with LSM in Figure 7B and for A1AT and ALGS in Supplemental Figures S11B, S12B, <http://links.lww.com/HC9/C100>. Informative proteins in BA and A1AT were mesenchymal, mesothelial, and endothelial origin, with macrophages also represented in A1AT (Figure 7C and Supplemental Figure S11C, <http://links.lww.com/HC9/C100>). Several proteins showed rank concordance across the 3 methods, including CHRD2L2*, EFEMP1*, TRIL, FRZB*, FBLN5*, and SEMA7A* for BA, and SEMA7A*, IL12AIEB13*, SEMA3G, HS6ST1, and

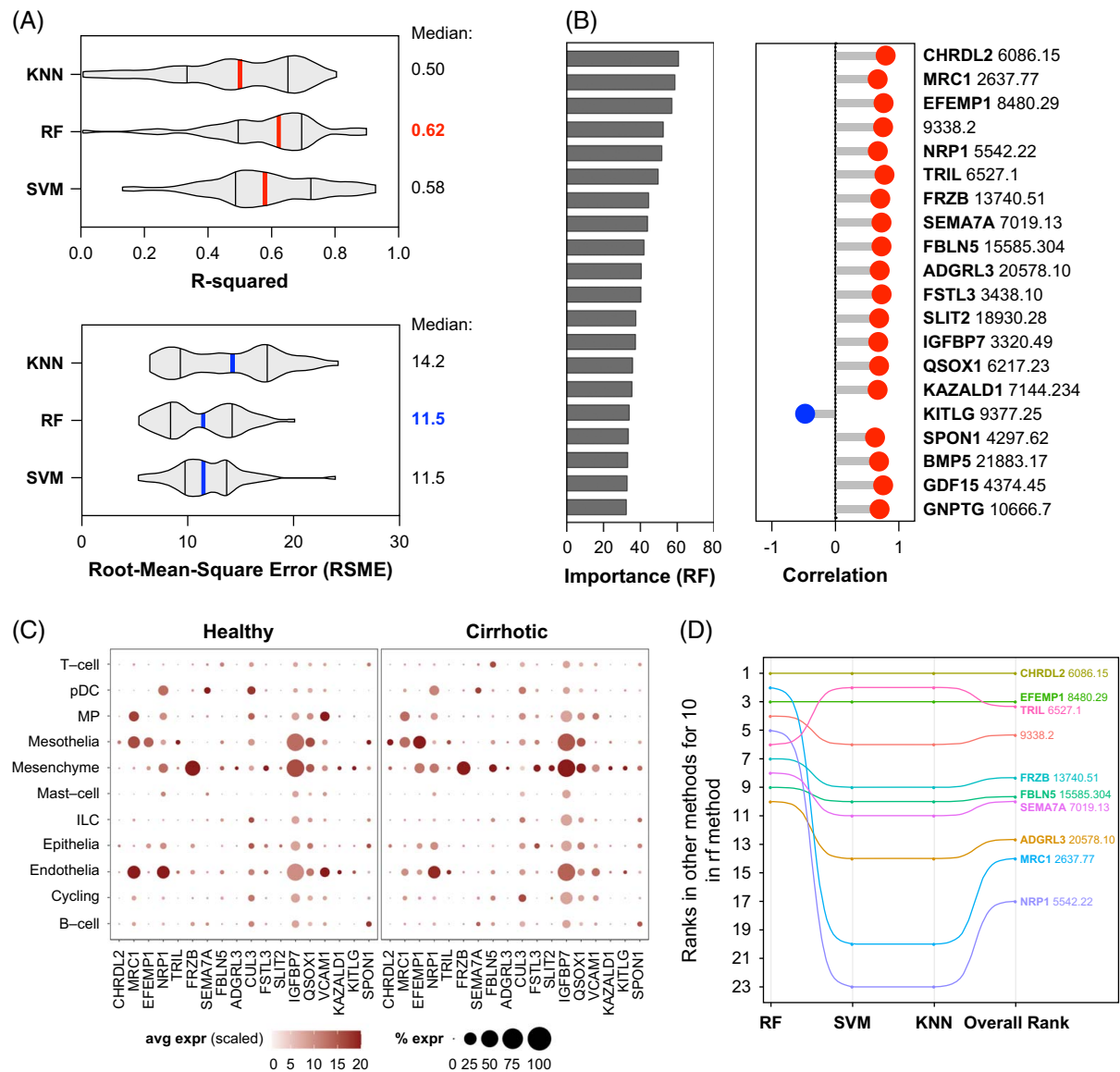


FIGURE 7 Machine learning regression of LSM in the BA. (A) Regression performance in RF, SVM, and KNN models using the squared Pearson correlation coefficient (R^2) and root mean square error (RMSE). (B) Top informative features from the RF method indicating feature importance and the Pearson correlation coefficient (R) with LSM. (C) Gene expression of the top 20 informative proteins (RF) in single-cell annotated cell types. (D) Rank concordance of the top 10 informative features across the 3 machine learning methods used. Abbreviations: BA, biliary atresia; KNN, k-nearest neighbor; LSM, liver stiffness measurement; RF, random forest; SVM, support vector machine.

NOTCH3* for A1AT (Figure 7D and Supplemental Figure S11D, <http://links.lww.com/HC9/C100>).

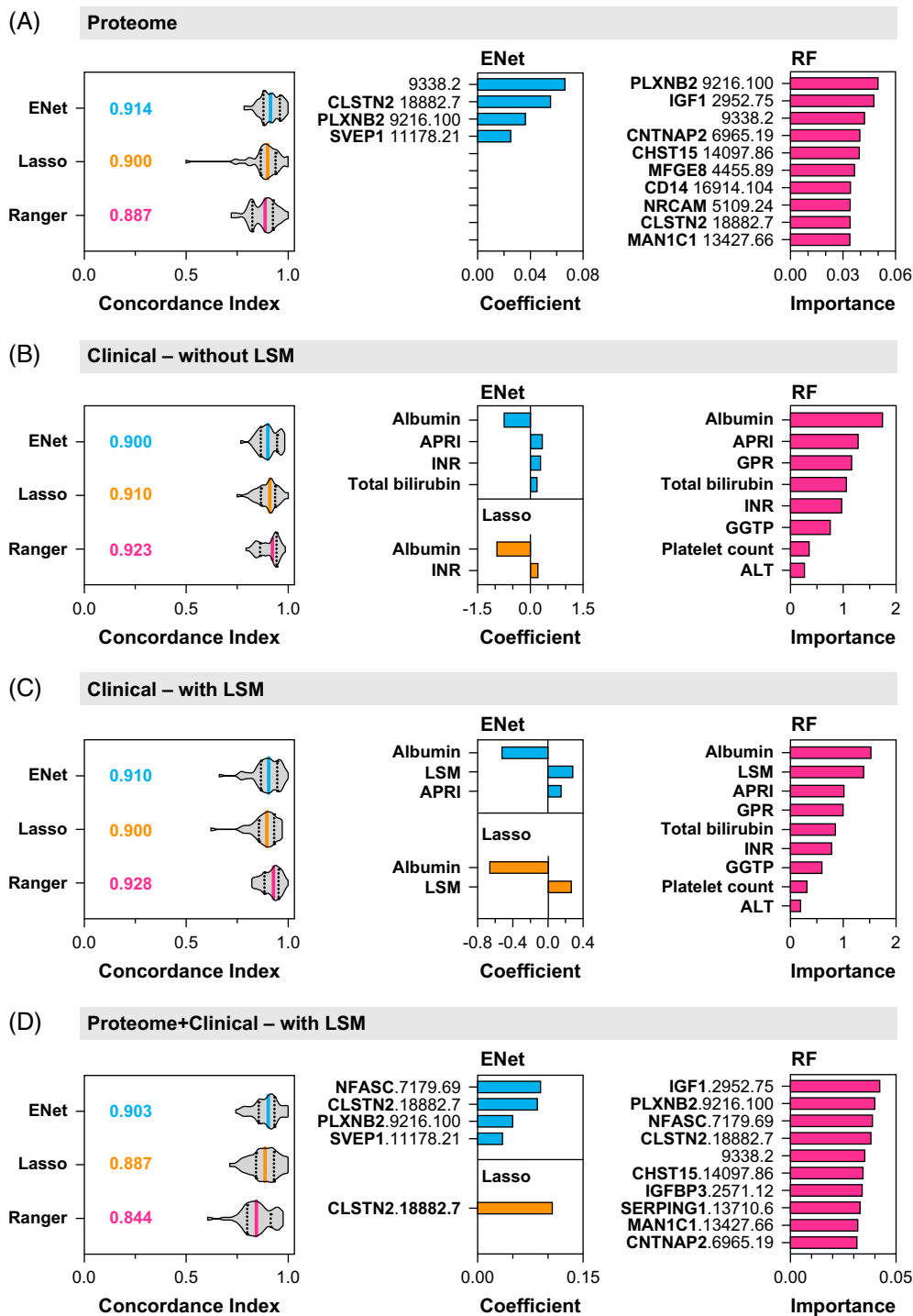
Prediction of time to transplantation

The ability of various parameters to predict outcome is depicted in Figure 8. The proteome was a strong predictor of outcome (Figure 8A), as were clinical parameters with or without LSM (Figures 8B, C). In the absence of LSM, the clinical elastic net model (Figure 8B) identified APRI, INR, and total bilirubin as essential features. LSM, albumin, and APRI were identified as critical clinical features when LSM was included (Figure 8C). When both

proteins and clinical parameters were used, predictive capacity was similar, and features from the proteome were called out as the most important predictors (Figure 8D). The similarity of the modeling is summarized in Figure 8E. While overall differences across predictor sets were minimal, particularly with lasso and elastic net, clinical data with LSM yielded slightly higher C-index values, suggesting a modest benefit of incorporating LSM.

DISCUSSION

A significant unmet need in pediatric hepatology is the identification of noninvasive markers of fibrosis and, by



(E)

Model Parameters	#variables	ENet		Lasso		RF	
		#features	C-index	#features	C-index	#features	C-index
Proteome	7596	4	0.914	0	0.900	300	0.887
Clinical – without LSM	8	4	0.900	2	0.910	8	0.923
Clinical – with LSM	9	3	0.910	2	0.900	9	0.928
Proteome+Clinical – with LSM	7605	4	0.903	1	0.887	273	0.844

FIGURE 8 Modeling of survival with native liver in BA. Concordance index and essential features from liver transplant-free survival analysis using random forest (ranger) and Cox models (ENet and lasso) based on (A) proteome, (B) clinical traits without LSM, (C) clinical traits including LSM, and (D) proteome and clinical traits including LSM. (E) Summary of outcome analysis. Abbreviations: APRI, aspartate aminotransferase to platelet ratio index; BA, biliary atresia; GGTP, gamma-glutamyl transpeptidase; GPR, gamma-glutamyl transpeptidase to platelet ratio; INR, international normalized ratio; LSM, liver stiffness measurement; RF, random forest.

corollary, markers of liver disease progression. Cholestatic liver diseases are distinct in pediatrics and contribute significantly to disease burden, including being the primary disease processes leading to liver transplantation in childhood. Fibrosis in diseases like BA can be profound, with cirrhosis developing in the first year of life in infants with persistent cholestasis. Our ability to accurately monitor and understand the biliary fibrosis pathogenesis in children is limited in large part by both the invasive nature of liver biopsy and the patchy nature of the fibrosis. The development of noninvasive elastographic techniques for assessing liver stiffness has the potential to transform approaches to monitoring children with chronic cholestasis. Unfortunately, elastography is not universally available and has high start-up costs. Its value in longitudinal assessment of pediatric cholestatic liver disease is uncertain. Biomarkers that correlate with liver stiffness have the potential to be especially useful. The current studies compare the elastographic assessment of liver stiffness with a powerful, unbiased plasma-based proteomic assay to further inform key questions in the monitoring and pathogenesis of pediatric cholestasis.

Three common causes of pediatric cholestasis were investigated: BA, ALGS, and A1AT. Liver stiffness in these disorders had been investigated by ChiLDReN in the FORCE study.^[3] In this study, samples were selected for analysis from participants who had valid baseline liver stiffness measurements and were representative of the range of portal hypertension states based upon CEPH criteria.^[9] Unlike in BA, there was a limited number of participants with ALGS or A1AT in the FORCE study with possible or definite CEPH. In general, the participants studied had compensated liver disease as noted by the clinical parameters elaborated in Figure 2. This population is representative of children with these cholestatic disorders followed in the outpatient setting. As with FORCE, there is disease specificity, which is not surprising given the differences in hallmark pathologic findings, especially in ALGS, where bile duct paucity is common. The plasma proteome, as shown in Figure 3, was similar between BA and A1AT, but markedly different, with many proteins having lower expression compared with ALGS. Pathway analysis of proteins positively correlated with LSM showed that the top-enriched pathway in all three diseases is EMT (Figure 4).

WGCNA module-based analysis provided further insights into correlations of the proteome and liver stiffness (Figure 5). In BA, the pink module was strongly correlated with LSM. A detailed examination of the pink

module yields several biologically interesting proteins (elaborated upon below) that are enriched in the EMT pathway. Hub genes identified using cytoHubba analyses suggest potential novel targetable proteins in pediatric fibrogenesis, including the Von Willebrand Factor. Proteins from 2 modules highly correlated with LSM in A1AT are enriched in EMT, IL6–JAK–STAT3 signaling, Apical junction, Complement, KRAS signaling, and related pathways (Supplemental Figure S5, <http://links.lww.com/HC9/C100>). WGCNA analyses of ALGS, as expected, called out distinct pathways and a different set of cells of origin (Supplemental Figure S6, <http://links.lww.com/HC9/C100>). Proteins found in the overlap of relevant modules in BA and A1AT (Figure 6) identified a very strong enrichment in the EMT, including 2 members (SLIT2* and THY1) of the portal fibroblast with mesenchymal stem cell features and their derived myofibroblast 7-gene signature.^[23] The second most prominent hallmark pathway was the apical junction, which is relevant for the cellular changes occurring during EMT.^[24] IPA identified axonal guidance signaling, which has been identified in mouse models of fibrosis and incorporates the SLIT-ROBO pathway, which is relevant in stellate cell activation, hepatic fibrosis, and cancer.^[25,26]

Using multiple machine learning approaches, specific elements of the plasma proteome were identified that were highly predictive of LSM in BA and A1AT (Figure 7 and Supplemental Figure S8, <http://links.lww.com/HC9/C100>). Predictive power was reduced for ALGS (Supplemental Figure S9, <http://links.lww.com/HC9/C100>). In BA, several interesting proteins were identified that were highly ranked in predicting LSM, including CHRDL2*, MRC1*, EFEMP1*, NRP1, TRIL, FRZB*, and SEMA7A*. The abundance of most of these proteins was markedly increased in participants with an LSM > 15 compared with healthy controls. Further investigation of these and other proteins as biomarkers of fibrosis and advancing cholestatic liver disease is clearly warranted. Ideally, one might develop a panel using these proteins that could replace elastography as an assessment of liver stiffness. ELISA-based assessments of VCAM1* and FRZB* were informative and support the possible use of more readily available assays (Supplemental Figure S6, <http://links.lww.com/HC9/C100>).

Together, these approaches identify fascinating groups of proteins for future investigation of the biology of advancing liver disease in pediatric cholestasis. Some of the proteins identified in a related proteomic

analysis linked to CEPH are represented in this analysis (eg, SEMA proteins and VCAM1*).^[7] Full curation of the interesting proteins identified in this analysis is beyond the scope of a single manuscript. As proof of principle, the biomedical relevance of several proteins that correlate with LSM to advanced liver disease is described. Slit homolog 2 protein (SLIT2*) is increased in primary biliary cholangitis, aggravates fibrosis after bile duct ligation, and is upregulated in diethylnitrosamine and thioacetamide-induced fibrosis.^[25,27] Semaphorin 7A (SEMA7A*) is increased in human fibrotic liver associated with hepatitis C or alcohol, increases in mouse bile duct ligation or CCl₄ treatment, and knocking it out reduces fibrosis in response to CCl₄.^[28] Serum level of insulin-like growth factor binding protein-7 (IGFBP7*) differentiates multiple grades of hepatic fibrosis in adults with chronic hepatitis C.^[29] Similarly, Semaphorin 3C (SEMA3C*) concentration in blood increases with advancing fibrosis stage in hepatitis C.^[30] EFEMP1* is a BA susceptibility gene that impacts elastic fiber formation and mechanical properties of the extrahepatic bile duct.^[31]

The biological relevance of proteins identified in these analyses is informed by understanding their potential cellular origins. We leveraged single-cell transcriptome data from adults with either cirrhosis or normal liver to gain insights into many of the identified proteins (Figures 5, 7 and Supplemental Figures S7–S9, <http://links.lww.com/HC9/C100>, S11, and S12, <http://links.lww.com/HC9/C100>).^[18] Macrophage, mesothelial, mesenchymal, and endothelial liver cell origins are called out in the proteins with the highest correlation with liver stiffness, especially in BA and A1AT. Many of the mesenchymal and endothelial-based proteins were increased in livers from adults with cirrhosis. In addition, the number of mesothelial cells in cirrhosis generally increased relative to that of the normal adult liver. In total, these findings, along with the pathway analyses, are strongly suggestive of the importance of the EMT pathway in hepatic fibrogenesis in these biliary diseases. EMT markers have been identified in the liver from patients with BA, with a suggestion of cholangiocytic origin.^[32–34] However, there remains conflicting evidence about the role of EMT in hepatic fibrogenesis.^[35] These studies provide further support for the pathophysiologic importance of EMT in progressive liver disease in childhood. Given the potential mesothelial origin of the proteins, these studies also implicate the relevance of the mesothelial–mesenchymal transition pathways in pediatric cholestatic liver disease.^[36] Ablation of mesothelin-positive activated portal fibroblasts reduced biliary fibrosis in mouse models, supporting the relevance of this pathway in cholestatic liver disease.^[37]

An important question is whether new biomarkers that predict outcomes can be derived from an unbiased analysis of the plasma proteome and its relationship to liver stiffness. This would be hypothesized since liver

stiffness predicts the outcome in biliary atresia.^[4] A key question for this study and the field is which outcome is the most relevant. In many studies, including this one, liver transplantation (LT) and/or time to LT is used as a sentinel outcome measure. This presumes that LT equates with a severe stage of progressive liver disease. That presumption may not be correct, as LT is performed in children with BA who have variceal hemorrhage or recurrent cholangitis and relatively compensated disease. In Alagille syndrome and progressive familial intrahepatic cholestasis, LT may be undertaken for intractable pruritus without decompensated liver disease. Liver transplantation is a choice, and its timing is dependent upon medical decision-making and liver allocation issues. For all these reasons, caution should be invoked when using LT as a major outcome for the study of pediatric liver diseases. Liver transplantation often preempts potentially fatal hepatic decompensation events, and as such, these endpoints are typically underpowered for clinical analyses of pediatric liver diseases. There is a critical and unmet need to develop a measure of liver disease severity that is continuous and applicable over the full natural history of pediatric liver diseases.

In this investigation, the plasma proteome performed as well as available clinical parameters in predicting time to LT. The addition of LSM and proteomic markers to clinical parameters did not enhance the ability to predict time to LT, although proteomic markers were of higher importance in the modeling. In this cohort, only 15 participants with BA came to LT in follow-up after the baseline LSM. Interestingly, in a related study, LSM was predictive of outcome in BA but did not add significantly to the predictive capacity of standard biochemical and hematologic markers of liver disease severity.^[4] The strong predictive power of routine clinical tests is valuable and makes the demonstration of superiority of another set of parameters difficult. The utility of serum albumin is fascinating. Serum albumin is a potential marker of nutritional status, hepatic synthetic function, and severity of portal hypertension. It is a strong predictor of the risk of variceal hemorrhage and of subsequent LT.^[38,39] Importantly, values in and near the normal range are predictive. Clinicians may need to be more attentive to small changes in serum albumin as important markers of progressive liver disease.

There are several important limitations in this study. As already stated, histologic correlates of disease progression were not surprisingly unavailable. Longitudinal analysis of changes in the proteome was beyond the scope of the current study. Prediction of outcome was restricted to time LT in BA, as there were insufficient events in participants with A1AT or ALGS. Cellular origin of the relevant proteins was derived from single-cell analyses with the assumption that they were liver-based. Future investigations of specific proteins will benefit from cell-specific investigations of normal

and diseased liver. The SomaScan assay in these studies is relatively expensive, which may necessitate adaptation to less expensive immune-based quantification of the abundance of selected proteins for routine clinical application of the findings. The exploratory ELISA investigations in this study support that potential approach.

In conclusion, there are rich opportunities to leverage the relationship between the plasma proteome and liver stiffness to develop noninvasive biomarkers of pediatric cholestatic liver disease severity, potentially in the form of a risk calculator. The disease specificity of the findings will need to be carefully considered. Globally, these findings provide potential mechanistic insights into fibrogenesis in these diseases, such as the strong enrichment of EMT in the pathway analyses. Novel therapeutic approaches may also emanate from these findings.

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CONFLICTS OF INTEREST

Kathleen Loomes consults and has received grants from Albeiro and Mirum. Ronald Sokol consults for Mirum Pharma and Ipsen/Albeiro. Evelyn Hsu received grants from Albeiro, Mirum, and Gilead. Jean Molleston received grants from Albeiro/Ipsen, Shire/Mirum, Gilead, and AbbVie. Philip Rosenthal consults and has received grants from Mirum and Ipsen. He consults for Taysha. Saul Karpen consults for Albeiro/Ipsen and Mirum. Simon Horslen consults for Ipsen/Albeiro. He received grants from Mirum. The remaining authors have no conflicts to report.

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